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Excellent. We will now **strictly use your new topology and configuration**:

- ✅ R1 G0/0 → Internet (DHCP, NAT outside)
- ✅ R1–R2 connected via **Serial S0/0/0**
- ✅ ISP domain: **100.10.10.0/30**
- ✅ LAN A: 192.168.10.0/24
- ✅ LAN B: 192.168.20.0/24
- ✅ Microservices from **Week03 (Phase 1 Mockup)** FastAPI architecture

We now reframe this as:

🧪 LAB 5 — Internet Edge + ISP Serial WAN + Week03 Microservices

🎯 LAB OBJECTIVE

Transform Week03 microservices into a **realistic enterprise deployment**:

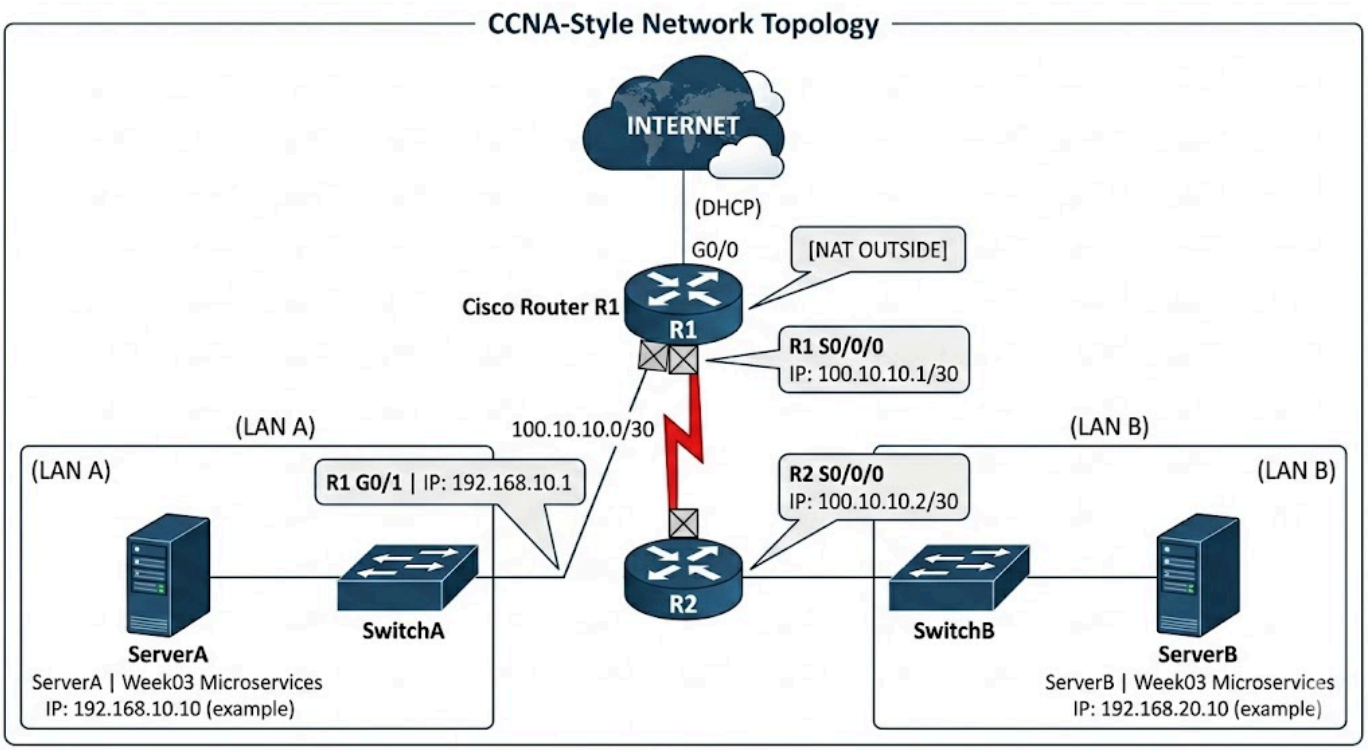
- R1 = Internet Edge Router (NAT + DHCP WAN)
- Serial WAN = ISP Domain (100.10.10.0/30)
- R2 = Branch Router
- Microservices deployed on ServerA and ServerB
- Validate cross-site microservice access over routed WAN + NAT

🗺️ LAB 5 TOPOLOGY (Reframed)

None

```
INTERNET
    |
(DHCP)
R1 G0/0
    |
[NAT OUTSIDE]
    |
```

```
                R1 S0/0/0
            100.10.10.1/30
                     |
                R2 S0/0/0
            100.10.10.2/30
                     |
          -----------------------
          |                      |
R1 G0/1 (LAN A)        R2 G0/1 (LAN B)
192.168.10.1            192.168.20.1
          |                      |
     ServerA                ServerB
Week03 Microservices   Week03 Microservices
```

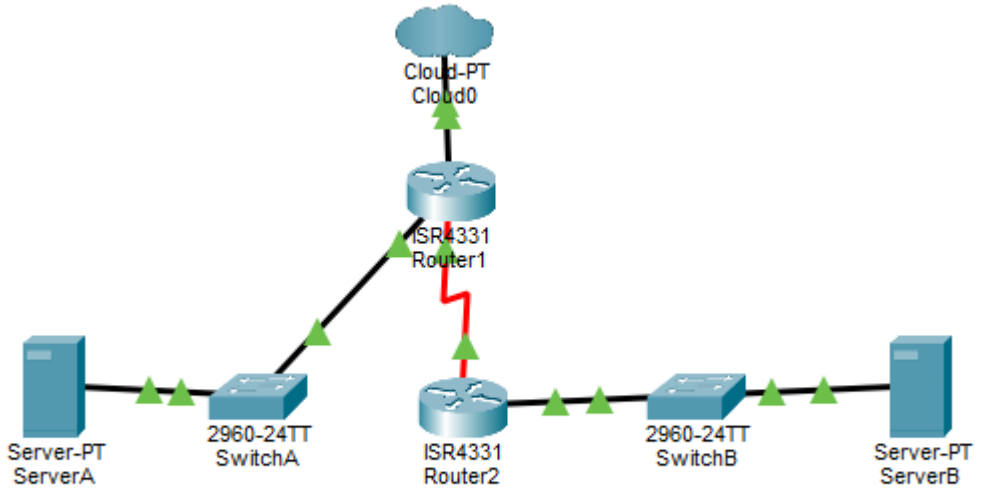


📍 ADDRESS TABLE (FINAL)

Device	Interface	IP	Role
R1	G0/0	DHCP	Internet

R1	S0/0/0	100.10.10.1/30	ISP
R1	G0/1	192.168.10.1/24	LAN A
R2	S0/0/0	100.10.10.2/30	ISP
R2	G0/1	192.168.20.1/24	LAN B
ServerA	eth0	192.168.10.10	Microservices A
ServerB	eth0	192.168.20.10	Microservices B

🔧 PART 1 — CISCO CONFIGURATION (STRICT)



🧱 R1 — Internet Edge Router

```
None
enable
conf t
hostname R1

interface g0/0
```

```
 ip address dhcp
 ip nat outside
 no shutdown

interface g0/1
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 no shutdown

interface s0/0/0
 ip address 100.10.10.1 255.255.255.252
 clock rate 64000
 no shutdown

ip route 192.168.20.0 255.255.255.0 100.10.10.2
ip route 0.0.0.0 0.0.0.0 g0/0

access-list 10 permit 192.168.10.0 0.0.0.255
ip nat inside source list 10 interface g0/0 overload

end
write memory
```

```
R1(config-if)#
R1(config-if)#interface g0/0/1
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ip nat inside
R1(config-if)# no shutdown

R1(config-if)#
R1(config-if)#interface s0/1/0
R1(config-if)# ip address 100.10.10.1 255.255.255.252
R1(config-if)# clock rate 64000
This command applies only to DCE interfaces
R1(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
R1(config-if)#
R1(config-if)#ip route 192.168.20.0 255.255.255.0 100.10.10.2
R1(config)#ip route 0.0.0.0 0.0.0.0 g0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
%Invalid interface type and number
R1(config)#
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R1(config)#ip nat inside source list 10 interface g0/0 overload
%Invalid interface number (Slot is empty)
R1(config)#
R1(config)#end
R1#write memory
Building configuration...
[OK]
R1#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to down

%SYS-5-CONFIG_I: Configured from console by console

R1#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
```

🧱 R2 — Branch Router

```
None
enable
conf t
hostname R2

interface g0/1
 ip address 192.168.20.1 255.255.255.0
 no shutdown

interface s0/0/0
 ip address 100.10.10.2 255.255.255.252
 no shutdown

ip route 192.168.10.0 255.255.255.0 100.10.10.1
ip route 0.0.0.0 0.0.0.0 100.10.10.1

end
```

```
write memory
```

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#
R2(config)#interface g0/0/1
R2(config-if)# ip address 192.168.20.1 255.255.255.0
R2(config-if)# no shutdown

R2(config-if)#
R2(config-if)#interface s0/1/0
R2(config-if)# ip address 100.10.10.2 255.255.255.252
R2(config-if)# no shutdown

R2(config-if)#
R2(config-if)#ip route 192.168.10.0 255.255.255.0 100.10.10.1
R2(config)#ip route 0.0.0.0 0.0.0.0 100.10.10.1
R2(config)#
R2(config)#end
R2#write memory
Building configuration...
[OK]
R2#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

R2#
```

💻 PART 2 — Deploy Week03 Microservices

Based on your Week03 architecture from `PROJECT_PROFILE.md`

We deploy:

- Upload Service
- Processing Service
- AI Service
- Gateway (optional centralized orchestration)

🛠️ On ServerA (LAN A)

```shell
cd automation/week03-microservices/phase1
python start_services.py
```

Services:

- Upload → 8000
- Processing → 8001
- AI → 8002
- Gateway → 9000

```
PS C:\Users\Viphu\Desktop\automation> cd .\week03-microservices\phase1\
PS C:\Users\Viphu\Desktop\automation\week03-microservices\phase1> python start_services.py
>>
🎯 Starting Phase 1 Microservices...
==============================================
🚀 Starting Upload Service on port 8000...
⏳ Waiting for Upload Service to become ready...
✅ Upload Service is ready!
🚀 Starting Processing Service on port 8001...
⏳ Waiting for Processing Service to become ready...
✅ Processing Service is ready!
🚀 Starting AI Service on port 8002...
⏳ Waiting for AI Service to become ready...
✅ AI Service is ready!
🚀 Starting Gateway Service on port 9000...
⏳ Waiting for Gateway Service to become ready...
✅ Gateway Service is ready!

🎉 All services are running!

🔍 Checking system health...
System status: healthy
  ✅ gateway: healthy
  ✅ upload: healthy
  ✅ processing: healthy
  ✅ ai: healthy

🧪 Running quick integration test...
📝 Created test file: test_data\sample.jpg
❌ Quick test failed: 422
   Response: {"detail":[{"type":"model_attributes_type","loc":["body","processing_options"],"msg":"Input should be a valid
dictionary or object to extract fields from","input":"{\"enable_processing\": true, \"processing_operation\": \"thumbnail\"
, \"enable_ai_analysis\": true, \"ai_analysis_type\": \"vision\"}"}]}
```

```
Available endpoints:
  Gateway:    http://localhost:9000
  Upload:     http://localhost:8000
  Processing: http://localhost:8001
  AI:         http://localhost:8002

Test commands:
  curl -X POST -F 'file=@test_data/sample.jpg' http://localhost:9000/process-file
  pytest tests/test_phase1_compliance.py -v

🔄 Press Ctrl+C to stop all services...

📊 Service Status:
----------------------------------------
Upload Service       🟢 Running
Processing Service   🟢 Running
AI Service           🟢 Running
Gateway Service      🟢 Running

📊 Service Status:
----------------------------------------
Upload Service       🟢 Running
Processing Service   🟢 Running
AI Service           🟢 Running
Gateway Service      🟢 Running

📊 Service Status:
----------------------------------------
Upload Service       🟢 Running
Processing Service   🟢 Running
AI Service           🟢 Running
Gateway Service      🟢 Running

📊 Service Status:
```

ServerA IP:

```
None
192.168.10.10
```

🛠️ On ServerB (LAN B)

Same deployment:

```shell
python start_services.py
```

ServerB IP:

```
None
192.168.20.10
```

```
PS C:\Users\Viphu\Desktop\automation> cd .\week03-microservices\phase1\
PS C:\Users\Viphu\Desktop\automation\week03-microservices\phase1> python start_services.py
🎯 Starting Phase 1 Microservices...
==================================================
🚀 Starting Upload Service on port 8000...
⏳ Waiting for Upload Service to become ready...
✅ Upload Service is ready!
🚀 Starting Processing Service on port 8001...
⏳ Waiting for Processing Service to become ready...
✅ Processing Service is ready!
🚀 Starting AI Service on port 8002...
⏳ Waiting for AI Service to become ready...
✅ AI Service is ready!
🚀 Starting Gateway Service on port 9000...
⏳ Waiting for Gateway Service to become ready...
✅ Gateway Service is ready!

🎉 All services are running!

🔍 Checking system health...
System status: healthy
   ✅ gateway: healthy
   ✅ upload: healthy
   ✅ processing: healthy
   ✅ ai: healthy

🧪 Running quick integration test...
📝 Created test file: test_data\sample.jpg
❌ Quick test failed: 422
   Response: {"detail":[{"type":"model_attributes_type","loc":["body","processing_options"],"msg":"Input should be a valid
dictionary or object to extract fields from","input":"{\"enable_processing\": true, \"processing_operation\": \"thumbnail\"
, \"enable_ai_analysis\": true, \"ai_analysis_type\": \"vision\"}"}]}

🎯 Phase 1 Microservices are ready!
```

🧪 PART 3 — WAN + INTERNET TESTING

🔵 Test 1 — Cross-Site Upload (LAN A → LAN B)

From ClientA:

```
None
curl http://192.168.20.10:8000/health
```

Should traverse:

ClientA → R1 → Serial ISP → R2 → ServerB

```
{"status":"healthy","service":"upload-service","timestamp":"2026-04-11T05:43:25.653750"}
```

🔵 Test 2 — Full Workflow Across ISP

From LAN A client:

```
None
curl -X POST http://192.168.20.10:9000/process-file
```

Flow:

Upload → Processing → AI → Response returned across WAN


```
{"detail":[{"type":"missing","loc":["body","file"],"msg":"Field required","input":null}]}
```

🔵 Test 3 — Internet Reachability

From ServerA:

```
None
ping 8.8.8.8
```

```
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=19ms TTL=113
Reply from 8.8.8.8: bytes=32 time=19ms TTL=113
Reply from 8.8.8.8: bytes=32 time=24ms TTL=113
Reply from 8.8.8.8: bytes=32 time=18ms TTL=113

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 24ms, Average = 20ms
```

Verify NAT overload working on R1:

```
None
show ip nat translations
```

🔎 Verification Commands

On R1

```
None
show ip route
show ip nat translations
show ip interface brief
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

R1#show ip nat translations
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.10.10.0/30 is directly connected, Serial0/1/0
L       100.10.10.1/32 is directly connected, Serial0/1/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/1
S    192.168.20.0/24 [1/0] via 100.10.10.2

R1#show ip nat translations
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES DHCP   up                    up
GigabitEthernet0/0/1   192.168.10.1    YES NVRAM  up                    up
GigabitEthernet0/0/2   unassigned      YES NVRAM  administratively down down
Serial0/1/0            100.10.10.1     YES NVRAM  up                    up
Serial0/1/1            unassigned      YES NVRAM  administratively down down
Vlan1                  unassigned      YES NVRAM  administratively down down
```

On R2

```
None
show ip route
ping 100.10.10.1
```

```
R2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 100.10.10.1 to network 0.0.0.0

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.10.10.0/30 is directly connected, Serial0/1/0
L       100.10.10.2/32 is directly connected, Serial0/1/0
S    192.168.10.0/24 [1/0] via 100.10.10.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/1
S*   0.0.0.0/0 [1/0] via 100.10.10.1

R2>ping 100.10.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.10.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 9/17/44 ms
```

🧠 Engineering Reflection (Week03 Integration)

From your Week03 profile :

This lab now demonstrates:

Concept	Where Demonstrated
Public vs Private Boundaries	R1 NAT edge
Distributed Services	ServerA & ServerB
Network Segmentation	Serial ISP /30
Horizontal Architecture	Two independent microservice stacks
Internet Edge Pattern	R1 as NAT Gateway
Service Isolation	LAN A vs LAN B

🔬 What Changed from Lab 4

Lab 4	Lab 5
10.10.12.0/30 simulated internet	Real ISP serial 100.10.10.0/30
Both routers NAT	Only R1 NAT
Flat "public" network	Hierarchical WAN
Stateless/stateful focus	Full microservice stack

🏗️ Architectural Significance

You now have:

- Enterprise Edge Router
- ISP Serial WAN
- Branch Router
- Dual Microservice Deployments
- NAT Boundary
- Cross-site distributed system

This is no longer a classroom NAT lab.

This is a **miniature cloud-edge + branch distributed architecture.**